

Substrate-Agnostic SERS Classification

Quick supervisor summary

May 6, 2026

Summary

The earlier SERS Siamese work was effectively a **chemical-substrate pair classifier**. The model learned classes such as `4np__AgNP`, `pyridine__PICO`, and `bt__pSERS`. That is useful for distinguishing measured pair conditions, but it does not establish substrate-agnostic detection.

The current work changes the task. The model now predicts **chemical identity only** and is evaluated by holding out an entire substrate family during training. This is the relevant test: if a substrate family is absent during training, can the model still identify the chemical on that held-out family?

Method	Accuracy	Bal. acc.	Macro F1	Weakest fold
Previous chemical-substrate Siamese evaluation	0.854	0.854	0.686	0.380
Best substrate-agnostic Siamese	0.975	0.975	0.977	Ag: 0.920
Best classical baseline	0.987	0.987	0.987	Ag: 0.960
Raw-spectrum Siamese control	0.440	0.440	0.399	Au: 0.000

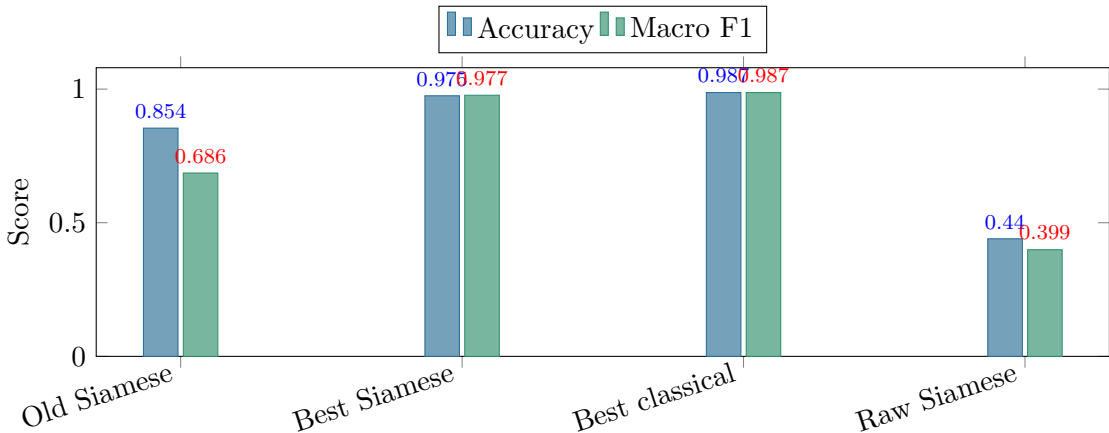


Figure 1: Main result comparison. For the grouped runs, macro F1 is computed over the true labels present in each held-out fold. The best substrate-agnostic Siamese model performs well, but the best non-deep baseline is slightly better on this small dataset. Raw spectra perform poorly, confirming that derivative preprocessing is important.

Dataset And Evaluation

The current substrate-agnostic evaluation uses 275 spectra from three chemicals: 4np, benzenethiol, and pyridine. The current matrix is:

Chemical	Ag	Au	PICO	pSERS
4np	25	0	25	25
benzenethiol	25	25	25	25
pyridine	25	25	25	25
n,n-dimethylformamide	0	228	0	0

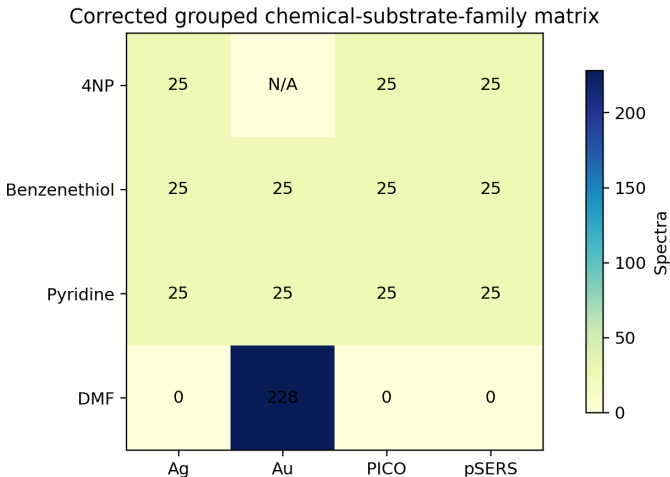


Figure 2: Current chemical-substrate-family matrix. DMF is present but only on Au, so it is not part of the current substrate-agnostic model.

Testing is **leave-one-substrate-family-out**. Each fold removes one full substrate family from training and tests on that held-out family:

Held-out family	Train	Test	Test chemicals
Ag	200	75	4np, benzenethiol, pyridine
Au	225	50	benzenethiol, pyridine
PICO	200	75	4np, benzenethiol, pyridine
pSERS	200	75	4np, benzenethiol, pyridine

This prevents direct memorization of the held-out substrate-family spectra. The remaining limitation is dataset size and independence: 25 spectra from one map are not equivalent to 25 independent preparations.

Preprocessing

The report notes a low-wavenumber device artifact below approximately 300 cm^{-1} . The current pipeline keeps only 330.906 to 1797.919 cm^{-1} , giving 663 spectral input points.

The best Siamese model uses first-derivative spectra:

1. crop the spectrum to $330\text{--}1800\text{ cm}^{-1}$;

2. apply per-spectrum standard normal variate normalization;
3. apply a Savitzky-Golay first derivative with window length 17 and polynomial order 3;
4. L2-normalize the final vector.

The best classical baseline uses the same idea but with the **second derivative**. The second derivative emphasizes peak curvature and suppresses broad baseline/background structure more strongly.

Best Siamese Model

The best substrate-agnostic Siamese model keeps the original metric-learning idea but changes the label and evaluation. It learns a 64-dimensional chemical embedding and classifies by nearest chemical prototype.

Component	Current setting
Input	First-derivative, SNV-normalized, L2-normalized SERS spectrum.
Encoder	Conv1D 1→16, ReLU, max pool; Conv1D 16→32, ReLU, max pool; flatten; linear layer to 64 dimensions; ReLU; L2 embedding normalization.
Loss	Triplet loss with margin 0.2.
Triplet sampling	Anchor and positive share chemical identity; cross-substrate positives are preferred. Negative has a different chemical; same-substrate negatives are preferred when available.
Training	100 epochs, batch size 32, Adam learning rate 10^{-3} , small Gaussian noise and spectral-shift augmentation. Training is run on GPU.
Inference	Embed training spectra, compute row-mean chemical prototypes, classify held-out spectra by nearest prototype.

Fold-level performance:

Held-out	Accuracy	Bal. acc.	Macro F1	Main error
Ag	0.920	0.920	0.919	6/25 4np predicted as benzenethiol .
Au	0.980	0.980	0.990	1/25 benzenethiol predicted as 4np .
PICO	1.000	1.000	1.000	None.
pSERS	1.000	1.000	1.000	None.

The saved CSV reports 0.660 macro F1 for the held-out Au fold because **4np** is not a true Au test class, but one spectrum is falsely predicted as **4np**. That default macro average includes the absent class and is not a fair fold-level summary. The table above reports macro F1 over the true labels present in each fold.

Best Classical Baseline

The best non-deep-learning baseline is:

second derivative + nearest centroid

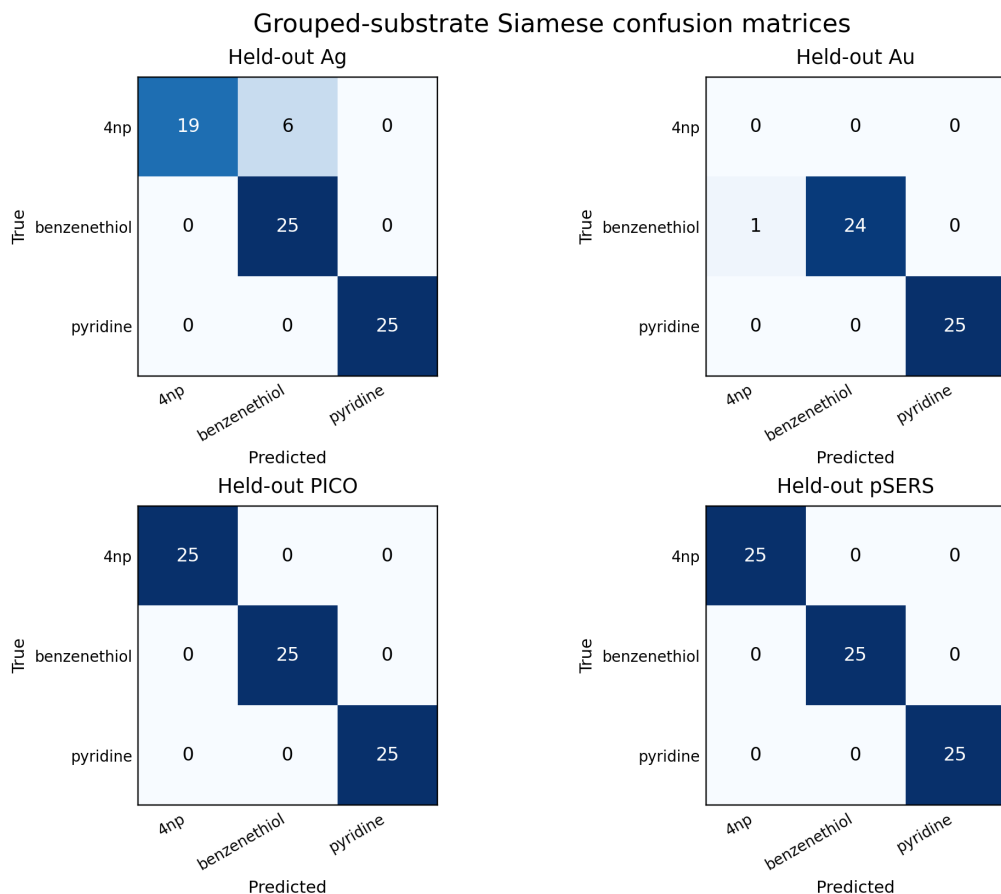


Figure 3: Best Siamese confusion matrices. The only meaningful failure mode is **4np** in the held-out Ag fold being pulled toward **benzenethiol**.

This model computes the second-derivative feature vector for every spectrum. During each leave-one-substrate-family-out fold, it computes the centroid of each chemical class from the training substrate families. A held-out test spectrum is assigned to the chemical whose centroid is closest in Euclidean distance.

Feature	Model	Accuracy	Bal. acc.	Macro F1
derivative_2	nearest centroid	0.987	0.987	0.987
derivative_1	nearest centroid	0.973	0.973	0.973
peak_emphasis	nearest centroid	0.963	0.963	0.963
derivative_2	cosine kNN, $k = 3$	0.950	0.950	0.946

This baseline is important because it is simpler than the Siamese model and slightly stronger on the current dataset. The spectra are already well organized in derivative space for these three chemicals. Future deep models should be compared against this baseline, not only against the old Siamese pair-classification results.

Geometry Evidence

The Siamese embedding improves chemical organization. Silhouette scores compare how close each spectrum is to its own group versus the nearest other group.

Space	Chemical silhouette	Substrate silhouette	Difference
Derivative input	0.304	0.101	0.203
Siamese embedding	0.797	-0.040	0.837

The embedding therefore clusters spectra much more by chemical identity and less by substrate family.

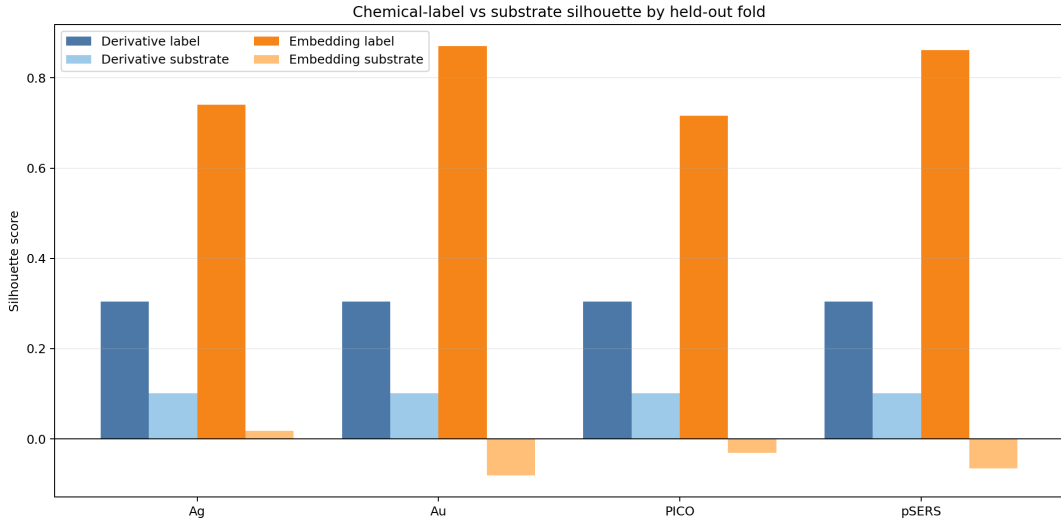


Figure 4: Chemical-label and substrate-family silhouette scores by held-out fold.

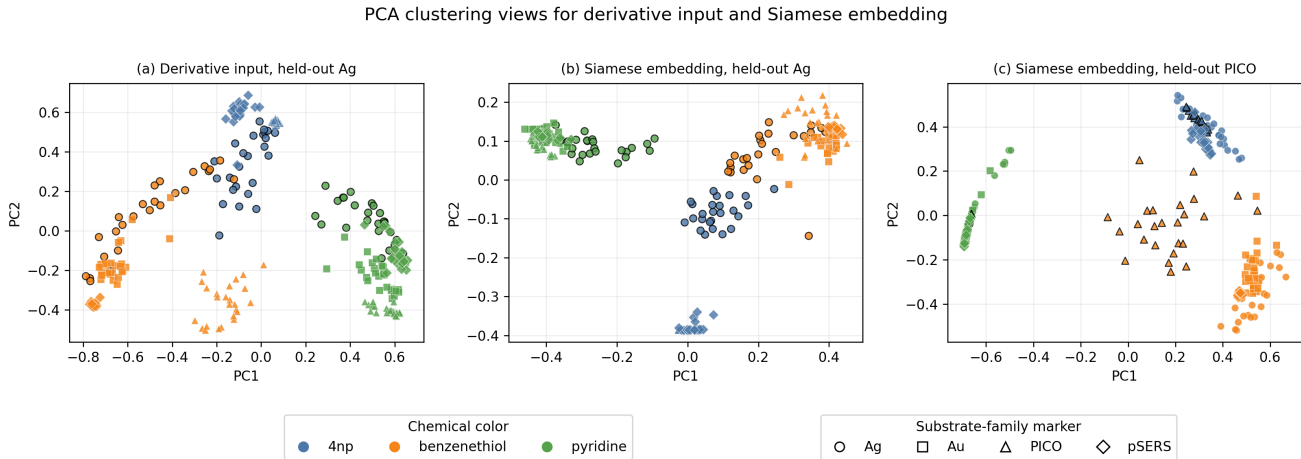


Figure 5: PCA clustering views. (a) Derivative-input PCA for held-out Ag. (b) Siamese-embedding PCA for held-out Ag. (c) Siamese-embedding PCA for held-out PICO. Color indicates chemical identity; marker shape indicates substrate family.

Interpretation And Next Steps

The current substrate-agnostic result is credible for the available three-chemical dataset, but it is not yet a broad SERS substrate-agnostic detector. The evidence is strongest for **benzenethiol** and **pyridine**, which are measured on all four substrate families. **4np** is valid on three substrate families but remains the main weak class in the held-out Ag fold.

The next dataset improvement should be to add another responsive chemical across all four substrate families. DMF is a candidate only if it responds reliably beyond Au.

The phrase "25 spectra per chemical-substrate-family cell" should be treated only as a short-term **minimum matrix-fill target**. It is enough to make a new cell usable in leave-one-substrate-family testing, but it is not a sufficient dataset size for robust deep learning. Siamese models help in small-data settings because they generate many pairs or triplets from limited spectra, but those pairs do not create new independent chemical or substrate variation.

For a stronger substrate-agnostic deep-learning claim, each new chemical-substrate-family cell should ideally include multiple independent preparations or maps, and preferably spectra collected across different sessions or days. More independent preparations are more valuable than simply collecting many correlated points from one map.

Bottom Line

We have moved from chemical-substrate pair classification to a true chemical-label, leave-one-substrate-family-out evaluation. The best Siamese model reaches 0.975 mean accuracy, and the best classical derivative baseline reaches 0.987. The current data support a strong proof of concept, but the next step for a stronger substrate-agnostic claim is another complete chemical across all substrate families.